Lecture Notes

Knowledge Graph Enhanced Educational Notes

1. Data Structure And Algorithm Subject

Definition:
The subject where stack and queue are covered in unit three.

1.1 Unit 3

Contains topics such as stack and queue.

- Stack: An abstract data type, commonly used in most programming languages. A linear data structure that follows a particular order (LIFO/FILO).
- Queue: A data structure topic covered in unit three.

Visual Representation:

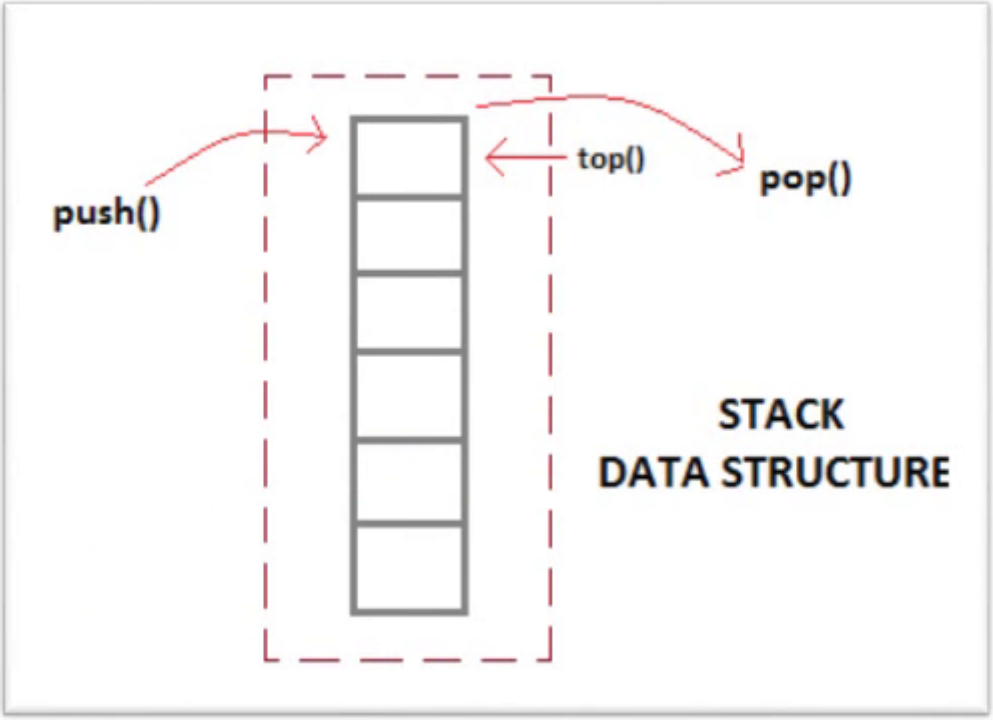

Figure 1: Diagram illustrating: The subject where stack and queue are covered in unit three.

2. Lifo (Last In First Out)

Definition:
The particular order in which operations are performed on a stack, where the last item added is the first one out.

- Related Concepts

- FILO (First In Last Out): An order that is the same as LIFO.

3. Filo (First In Last Out)

Definition:
An order that is the same as LIFO.

4. Top Of The Stack

Definition:

The location where elements are added to and removed from; the most recently added items are placed here.

Visual Representation:

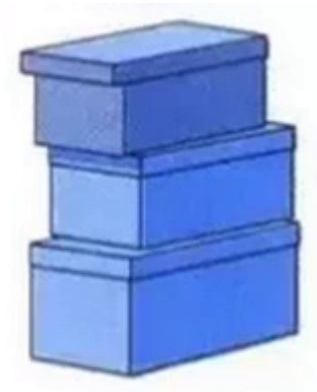

Figure 4: Diagram illustrating: An abstract data type, commonly used in most programming languages. A linear data structure that follows a particular order (LIFO/FILO).

5. Stack Implementation Methods

Definition:

Ways to implement a stack, including by means of Array, Structure, Pointer, and Linked List.

6. Important Questions (Sppu Exam)

Definition:

Frequently asked questions in the SPPU exam pattern, covering stack definition, implementation, implicit/explicit stack, recursion importance, and applications of stack.

- Related Concepts

- Stack: An abstract data type, commonly used in most programming languages. A linear data structure that follows a particular order (LIFO/FILO).
- Stack Implementation Methods: Ways to implement a stack, including by means of Array, Structure, Pointer, and Linked List.
- Types of Stack: Mainly two types: implicit stack and explicit stack.
- Recursion: A functionality implemented by the compiler using the implicit stack; also relevant to explicit stack algorithms and an important question topic.
- Applications of Stack in Data Structure: Includes converting infix expression to postfix and prefix, evaluating postfix expressions, reversing strings, checking well-formed parentheses, processing subprogram/function calls, parsing, decimal to binary conversion, and backtracking algorithms.

7. Recursion

Definition:

A functionality implemented by the compiler using the implicit stack; also relevant to explicit stack algorithms and an important question topic.

8. Stacks

Definition:

A new chapter introduced in the presentation.

- Related Concepts

- Definition of Stack: What a stack is all about.
- Glass Jar Analogy: A way to imagine a stack where objects are placed inside, making it easier to visualize restrictions.

Visual Representation:

Figure 8: Diagram illustrating: An abstract data type, commonly used in most programming languages. A linear data structure that follows a particular order (LIFO/FILO).

9. Presentation

Definition:

Introduces the concepts of stacks.

- Components and Structure

- Outline of Presentation: The outline of this presentation.

- Related Concepts

- Stacks: A new chapter introduced in the presentation.

10. Definition Of Stack

Definition:

What a stack is all about.

- Related Concepts

- Linear Data Structure: All the data or elements are arranged in a sequential manner or in linear fashion.
- Insertions: Allowed only at the end called the top of the stack in a stack.
- Deletions: Allowed only at the end called the top of the stack in a stack.

11. Real Life Examples Of Stacks

Definition:

Some real life examples of stacks.

- Components and Structure

- Stack of Books: A real life example of stacks.
- Stack of Coins: A real life example of stacks.

12. Stack As An Adt

Definition:

What is a stack as an ADT (Abstract Data Type).

- Related Concepts

- Abstract Data Type (ADT): Means stack uses different types of operations, whose theory allows direct use of push and pop operations during implementation.
- User Point of View: The perspective from which stack operations are considered when defining a stack as an ADT.
- Implementation Details: How stack operations are performed, which is not of interest when defining a stack as an ADT from a user's perspective.

13. Stack Operations

Definition:

Primary and secondary operations supported in case of stack.

- Related Concepts

- Primary Operations: A category of stack operations.
- Secondary Operations: Another category of stack operations.

14. Insertions

Definition:

Allowed only at the end called the top of the stack in a stack.

- Related Concepts

- Top of the Stack: The location where elements are added to and removed from; the most recently added items are placed here.

15. Deletions

Definition:

Allowed only at the end called the top of the stack in a stack.

- Related Concepts

- Top of the Stack: The location where elements are added to and removed from; the most recently added items are placed here.

16. Array

Definition:
A type of linear data structure, like stack or linked list.

17. Linked List

Definition:
A type of linear data structure, like stack or array.

18. Access Object

Definition:
To retrieve an object, usually the top most one in a stack example.

19. Add Object

Definition:
To place an object, usually at the top most place in a stack example.

20. Glass Jar Analogy

Definition:
A way to imagine a stack where objects are placed inside, making it easier to visualize restrictions.

- Related Concepts
- Closed at Bottom: A characteristic of the glass jar in the analogy, implying no access from below.
- Opened at Top: A characteristic of the glass jar in the analogy, implying access from above.
- One Way to Insert/Delete: The singular method for modifying objects in a stack (from the top).

21. Closed At Bottom

Definition:
A characteristic of the glass jar in the analogy, implying no access from below.

22. Opened At Top

Definition:
A characteristic of the glass jar in the analogy, implying access from above.

23. One Way To Insert/Delete

Definition:
The singular method for modifying objects in a stack (from the top).

24. User Point Of View

Definition:

The perspective from which stack operations are considered when defining a stack as an ADT.

25. Implementation Details

Definition:

How stack operations are performed, which is not of interest when defining a stack as an ADT from a user's perspective.

26. Top Operation

Definition:

A secondary stack operation which simply returns the last inserted element without removing it.

- Related Concepts

- Top of the Stack: The location where elements are added to and removed from; the most recently added items are placed here.

27. Size Operation

Definition:

A secondary stack operation which simply returns the size or the number of elements in the stack.

28. List Of Commands

Definition:

A sequence of stack commands to be executed.

- Components and Structure

- Push(1): A command to push the element '1' inside the stack.
- Push(2): A command to push the element '2' inside the stack.
- Push(3): A command to push the element '3' onto the stack.
- Pop(): A command to delete the top most element from the stack.
- IsFull(): A command to check if the stack is full.